STEEL DEFECT DETECTION

Cocosila Rares

Manole Catalin

PREZENTARE MIDTERM

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SCOPUL PROIECTULUI

- Localizarea si clasficarea defectelor pe suprafete metalice
- La final, dorim sa prezentam o gama de modele care poate verifica si clasifica kilometrii de ben zide metalurgii si vom specifica ce model te va ajuta cel mai mult in functie de resursele tale.

STRUCTURA DATELOR

- Severstal: Steel Defect Detection (Kaggle) 12.6k - Training | 5506 - Testing

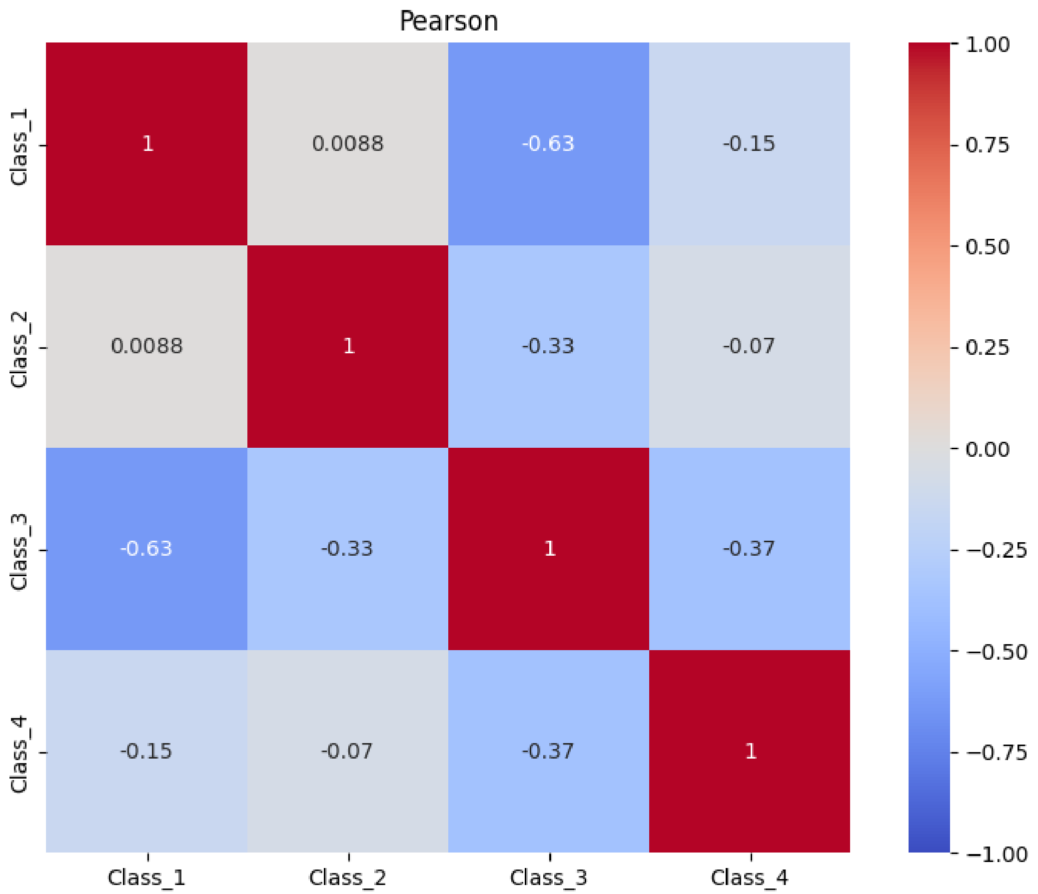
Fiecare imagine poate avea un defect, mai multe defecte, sau niciunul (defectele sunt segmentate in clasa "`ClassId = [1, 2, 3, 4]`").

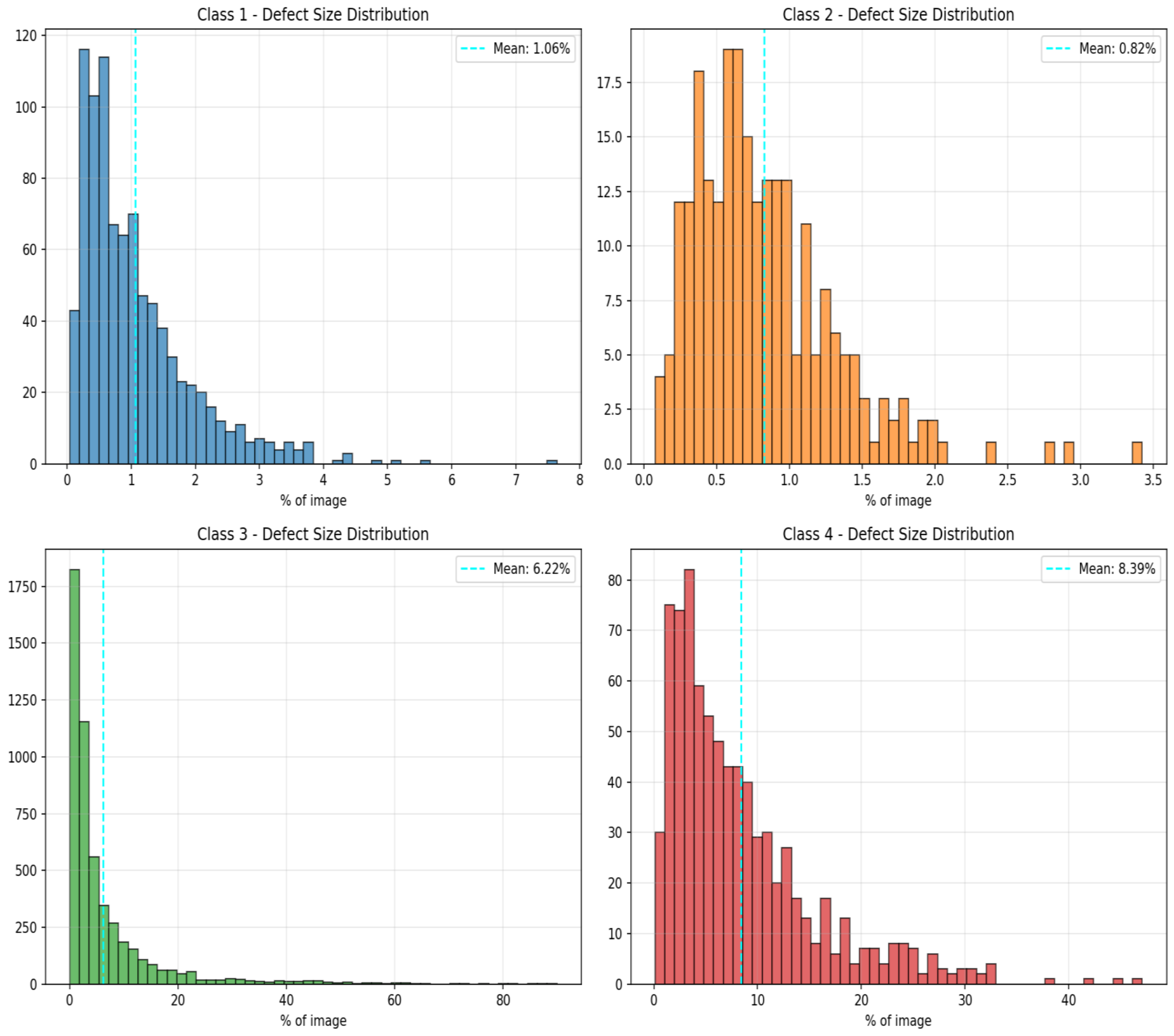
`Segmentul fiecare clase este encodat intr-un singur rand.`

- NEU surface defect database 1,440 - Training | 360 - Testing

// UNDER CONSTRUCTION //

ANALIZA DATELOR





MODELE

1. U-Net - "U-shaped" encoder-decoder structure with skip connections
2. Mask R-CNN - Region-based CNN which divides object detection task into two stages: region proposal and object classification
3. YOLOv11- Popular object detection & image segmentation model
4. Vit - Deep learning architecture that applies the Transformer model to images
5. Sam – Model developed by Meta; transformer-based encoder and a mask decoder that includes convolutional layers to produce segmentation masks.
6. Dino - self-supervised Vision Transformer Model

EVALUARE (METRICS)

- Intersection over Union (IoU)

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} = \frac{TP}{TP + FP + FN}$$

- Dice Coefficient

$$Dice = \frac{2 \times \text{Area of Overlap}}{\text{Total number of pixels in both}} = \frac{2TP}{2TP + FP + FN}$$

- Pixel Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

METODE DE OPTIMIZARE

- **Mixed Precision (AMP)**
  - using both 16-bit and 32-bit floating point numbers during training
  - Faster training, less memory usage
- **Fast Lookups**
  - caching data using dict. or index lists
  - time complexity reduced
- **Early Stopping**
  - stop training if the loss stops improving
  - avoids overfitting

REZULTATE SI RECOMANDARI

- **// UNDER CONSTRUCTION //**

MAI DEPARTE

- Optimizari pentru imbunatatirea modelelor
- Vizualizari de predictii
- Experimentarea parametrilor

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- Vit: https://github.com/google-research/vision_transformer
- Sam: https://github.com/facebookresearch/segment-anything?utm_source=chatgpt.com
- Dino: https://github.com/facebookresearch/dino?utm_source=chatgpt.com